



Application Descriptions

7

Energy Management

80

Introduction

Summary

This document introduces the KNX Energy Management Application Domain.

This document WGI Approved.

Document updates

Version	Date	Modifications
1.0	2024.12.13	new
1.1	2024.12.18	Editorial changes
1.1	2025.02.11	Font Changed to Arial 10, Figure 3 updated
01.01.01	2025.06.10	Creation of the WGI Approved version.

References

None.

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1 Application Domain Energy Management

Energy Management as part of a comprehensive Building Management System controls and optimizes the use of energy for operation of a home or building.

Energy Management is needed since renewable energies such as wind and sun are not permanently available and storing energy is limited and expensive. Therefore, attempts must be made to use the energy when it is available.

Energy management starts with integrated control of lighting, solar protection and heating, ventilation and air conditioning (HVAC) in building spaces. This integrated room control provides the desired comfort level of users in the space while minimizing the energy usage for the space (room).

Functions used for integrated room control are defined in the KNX Specifications, Volume 7:

- Part 7/10 “HVAC General Functional Blocks”
- Part 7/11 “Hot water heating”
- Part 7/13 “Terminal Unit Functional Blocks”
- Part 7/14 “Ventilation, Air Conditioning, and Cold Water”
- Part 7/20 “Lighting”
- Part 7/50 “Shutters and Blinds”

Those Parts cover the emission of energy in a space.

This document describes the source/sink, conversion, storage and distribution functional blocks for energy management in a building.

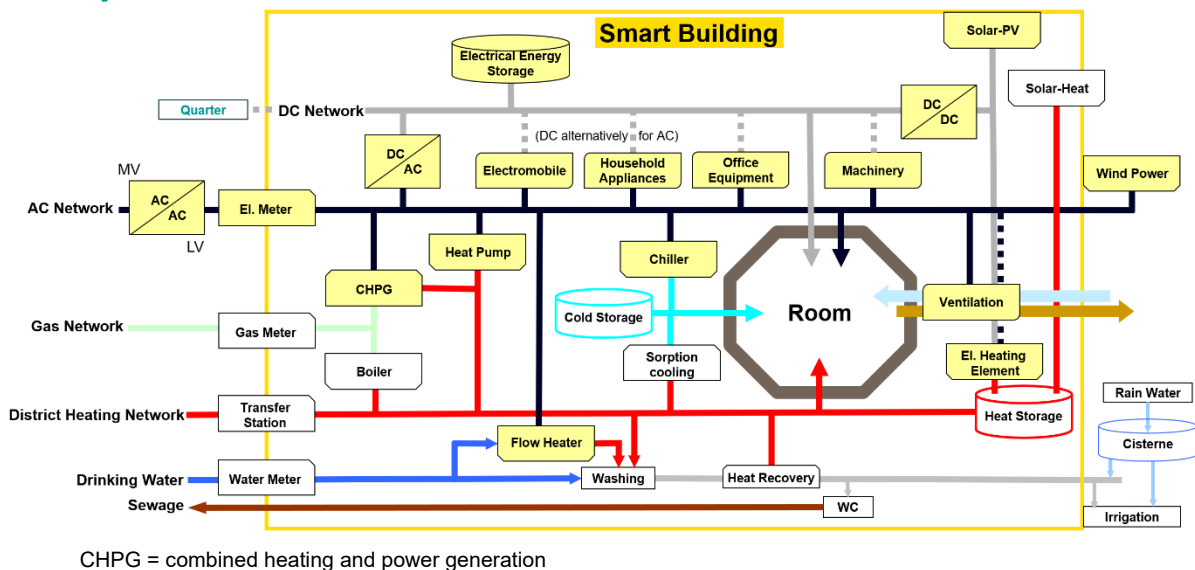


Figure 1 - Utility infrastructure in Buildings

The room receives energy via different networks for AC power, DC power, cold water, hot water / heating, supply and exhaust air ducts (figure 1).

These different utility networks connect specific source, conversion, storage and processing elements in the utility infrastructure of a building.

Source elements can be Solar-PV, Solar-Heat, Wind Power, Heat Recovery, Boiler

Conversion elements can be DC/AC inverter, DC/DC inverter

Storage elements can be Electrical energy storage, Heat storage, Cold storage

Processing elements can be Electro mobile, Household Appliances, Office Equipment, Machinery, Heat Pump, Combined Heating and Power Generation (CHPG), Flow Heater, Chiller, Sorption Cooling, Ventilation, Electric Heating Element, Heat Recovery

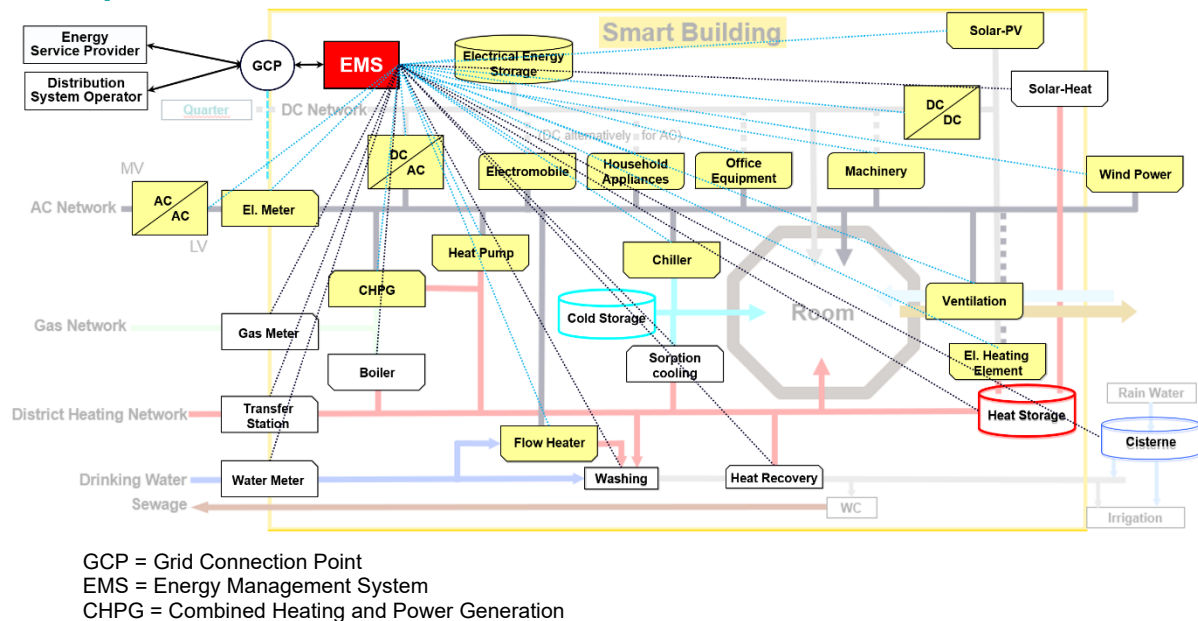


Figure 2 - Utility Infrastructure in Buildings with Energy Management System

The Energy Management System (EMS) controls the different source, storage and processing elements to optimize energy supply and usage in the building. Part of this optimization is to control the energy and water sourced from utility suppliers through electrical power, gas, district heating / cooling and water distribution networks. Electrical power may be supplied from the building to the electrical distribution network.

This document defines Functional Blocks of different energy source, storage and processing elements.

The Functional Blocks are divided in 3 groups, which are applications under the application domain “Energy Management”.

1. Photovoltaics
2. HVAC for EMS
3. E-mobility

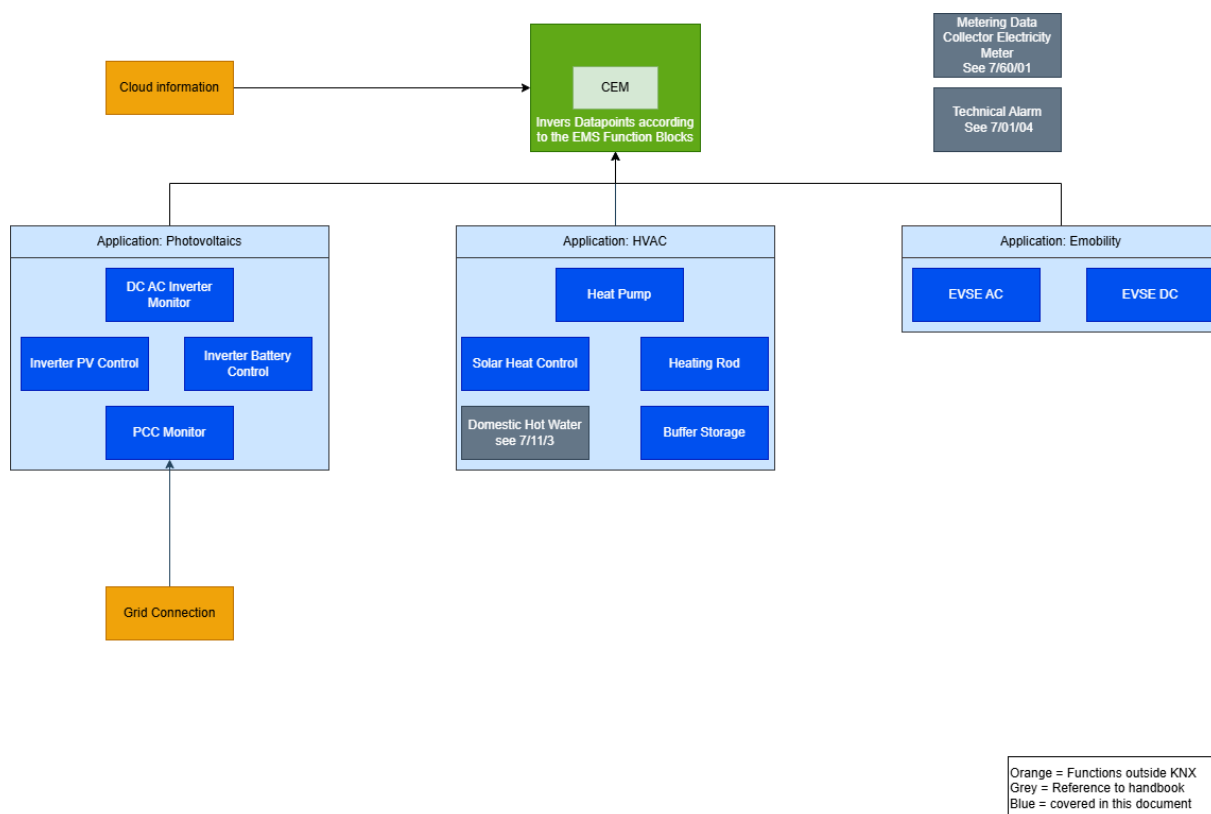


Figure 3 - Functional Blocks concerning the Application Domain EMS

The Functional Blocks in blue are Functional Blocks specific for energy management.

The Functional Blocks in grey are Functional Blocks of other Application Domains.

The orange rectangles are additional (non KNX) inputs in the EMS.